The Infinite Thread Theory: A Scale-Invariant Framework for Information Conservation and Field Collapse

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Abstract

This paper introduces an exploratory theoretical framework—the Dynamic Network Alignment Model (DNAM)—to investigate scale-invariant vector contributions within a network topology. Traditional models often introduce denominator-driven boundary singularities when modeling friction. I propose a multiplier-gated attenuation coefficient that maps conflicting internal parameters using linear scalar summation, ensuring numerical stability as variables approach zero. The framework's internal consistency and threshold dynamics are explored using a computational simulation app, demonstrating how stable crossover states can be modeled under closed mathematical conditions.

1 Introduction & Theoretical Integration Framework

Modern physics remains deeply fragmented by an artificial boundary separating the quantum realm from the classical macroverse. The establishment relies on two mutually exclusive equation sets: probabilistic wave mechanics for subatomic scales, and deterministic geometric spacetime profiles for macroscopic systems. This paper introduces a scale-free network blueprint exploring whether the underlying laws governing energy conservation and structural field realignment can be modeled identically across diverse regimes.

2 Scope and Empirical Status

The Infinite Thread Theory presented herein functions as an independent, exploratory theoretical framework proposing a scale-invariant model of information conservation, coherence, field interactions, and cyclical energy redistribution. This document includes targeted mathematical formulations and accompanying computational experiments intended specifically to explore the structural implications of the framework's rules.

It is critical to distinguish between the internal logical consistency of these equations, the deterministic outcomes of the programmatic sandbox, and actual observations of the physical universe. The simulation demonstrates the precise behavior of the programmed rules, verifying that the model behaves as intended under closed conditions; it does not, by itself, demonstrate that these rules describe physical processes in nature. This theory is a work in progress and has not yet been empirically validated or established as a physical theory. It is presented to invite rigorous discussion, academic critique, and the eventual development of falsifiable physical testing methods.

3 The First Axiom: Scattering-Based Coherence Decay Hypothesis

The framework models the time-evolution of a propagating signal's coherence within a scattering environment. Let $Q(t)$ represent a purity-inspired scalar state variable tracking system coherence. The model hypotheses that interaction with an uncalibrated environment reduces this value toward its mixed-state limit over time. To avoid non-physical excursions below the formal quantum boundary for a two-level system ($1/2 \leq Q \leq 1$), the environmental relaxation decay is represented by:

$$\frac{dQ}{dt} = -\Gamma_Q \left(Q - \frac{1}{2} \right), \quad Q(t) = \frac{1}{2} + \left(Q_0 - \frac{1}{2} \right) \exp(-\Gamma_Q t) \tag{1}$$

Where the effective decay rate Γ_Q is derived from the standard collision frequency of kinetic theory, modified by an interaction efficiency parameter:

$$\Gamma_Q = n \langle \sigma(v_{\text{rel}}) v_{\text{rel}} \eta(v_{\text{rel}}) \rangle \tag{2}$$

Here, n denotes the scattering target number density (m^{-3}), $\sigma(v_{\text{rel}})$ is the velocity-dependent scattering cross-section (m^2), v_{rel} isolates the relative speed between the signal and targets (m/s), and the dimensionless factor $\eta(v_{\text{rel}})$ represents a coupling efficiency hypothesis. Within the wider framework, this standalone decay profile functions as the limiting non-pushed condition where active collective pumping drives are absent, explicitly establishing the baseline dissipative floor for the system noise metrics.

4 The Phase Space Alignment Axiom

The model explores how individual node contribution vectors combine within a collective network configuration. Let each node i within an N-node system be modeled as a two-level quantum system (qubit) whose density matrix is mapped uniquely to a polarization spin vector $\vec{s}_i$ inside the Bloch sphere. The purity of each state is structurally bound to the length of its spin vector: $\|\vec{s}_i\| = \sqrt{2Q_i - 1}$.

The net collective alignment vector $\vec{R}$ projected by the network configuration onto the localized background field is calculated via the linear summation of these individual spin vectors:

$$\vec{R} = \sum_{i=1}^{N} w_i \vec{s}_i \tag{3}$$

Where w_i is a non-negative assigned weight. For cases where individual Bloch vectors are non-zero, this can be expressed using directional unit vectors $\vec{P}_i = \vec{s}_i / \|\vec{s}_i\| as \vec{R} = \sum w_i \sqrt{2Q_i - 1}\, \vec{P}_i$. Decoupled from the vector direction, a dimensionless Normalized Coherence Scalar C tracks the total global alignment of the system:

$$C = \begin{cases} \frac{\|\vec{R}\|}{\sum_{i=1}^{N} w_i \sqrt{2Q_i - 1}} & \sum_{i=1}^{N} w_i \sqrt{2Q_i - 1} \neq 0 \\ 0 & \sum_{i=1}^{N} w_i \sqrt{2Q_i - 1} = 0 \end{cases} \tag{4}$$

The fallback value $C = 0$ serves as a defined mapping convention for the isotropic mixed-state or zero-weight case, rather than representing an independent physical measurement of alignment. The denominator vanishes when all weighted Bloch-vector magnitudes vanish, which incorporates cases where specific tracking weights are assigned to zero.

5 The Principle of Bounded Volumetric Energy Fields

The framework proposes that localized network sectors accumulate potential within a finite control volume until a critical energy density condition initiates a non-linear state transition. Let $u(t) = U(t)/V$ represent the localized energy density measured in Joules per cubic meter ($\mathrm{J/m^3}$). By applying the quotient rule, the time-evolution of this density across a variable volume boundary is modeled by:

$$\frac{du}{dt} = \frac{P_{\text{in}} - P_{\text{loss}}}{V} - \frac{u}{V}\frac{dV}{dt} \tag{5}$$

Where P_{in} and P_{loss} track all net power transfers (J/s or Watts) crossing the control volume boundary. In a moving or deforming control volume, boundary transport terms and mechanical boundary work must be included consistently to form a complete energy-balance profile. For a fixed system geometry ($dV/dt = 0$), this simplifies directly to an energy-balance equation:

$$\frac{du}{dt} = \frac{P_{\text{in}} - P_{\text{loss}}}{V} \tag{6}$$

An instability relaxation cascade is triggered dynamically when the state density clears a designated threshold parameter:

$$u(t) \geq u_{\text{trigger}}, \quad u_{\text{trigger}} < u_{\text{critical}} \tag{7}$$

Here, u_{critical} functions as a theoretical parameter representing the upper asymptotic ceiling of the charging profile rather than an unconstrained physical bound.

6 The Co-Evolutionary Feedback Loop

The framework explores a phenomenological model where localized field energy density and node quantum purity undergo a coupled feedback loop. Let $u(t)$ represent the universal stored energy density ($\mathrm{J/m^3}$), and let P_{aligned} denote the non-negative rate of power transfer (J/s) channeled via coherent node alignment: $P_{\text{aligned}} = \zeta \max(0, \vec{R} \cdot \vec{B}_{\text{local}})$, where ζ carries appropriate coupling dimensions. The time-evolution of the stored field density follows a standard relaxation equation:

$$\frac{du}{dt} = \alpha \cdot Q \cdot (u_{\text{critical}} - u) \tag{8}$$

Where α represents a characteristic relaxation frequency ($\mathrm{s^{-1}}$). Under a constant, positive net input power regime across a fixed volume V, this relaxation frequency maps directly to the simulator constants via the structural identity $\alpha = (P_{\text{in}} - P_{\text{loss}})/(V \cdot u_{\text{critical}})$. Concurrently, the evolution of state purity $Q(t)$ is modeled as a competition between an engineered dissipation pump and an environmental noise loss term:

$$\frac{dQ}{dt} = \beta_{\text{gain}} \cdot P_{\text{aligned}} \cdot (1 - Q) - \gamma_{\text{loss}} \cdot \left(Q - \frac{1}{2}\right) \tag{9}$$

Where γ_{loss} is the environmental decoherence rate ($\mathrm{s^{-1}}$), and β_{gain} represents a coupling coefficient ($\mathrm{J^{-1}}$), keeping the interval $1/2 \leq Q \leq 1$ mathematically invariant. Within computational applications, the expression for P_{aligned} is treated as a reduced proxy model defined as $\max(0, \Lambda P_{\text{in}})$, which serves as a phenomenological placeholder rather than an explicit vector-field calculation loop.

7 The Physical Analogy of the Ink Drop

he framework models the dispersal of phase information into unobserved environmental degrees of freedom as analogous to a drop of ink diffusing within a closed vessel of fluid. Let $c(\vec{x}, t)$ represent the local concentration ($\mathrm{m^{-3}}$) of an informational tracer across a coordinate vector $\vec{x}$. Microscopic mass conservation is governed by the standard diffusion equation:

$$\frac{\partial c(\vec{x}, t)}{\partial t} = \nabla \cdot (D \nabla c(\vec{x}, t)) \tag{10}$$

Where D is the spatially dependent diffusion tensor. Under zero external flux conditions across the boundary walls, the global integrated tracer mass remains invariant:

$$\frac{d}{dt} \int_V c(\vec{x}, t), dV = 0 \tag{11}$$

In classical statistical mechanics, fine-grained phase-space volume is conserved under Liouville's theorem, keeping fine-grained Gibbs entropy invariant. Under suitable coarse-graining and mixing assumptions, unresolved fine-scale structure can produce an effective increase in coarse-grained entropy, even while fine-grained Gibbs entropy remains invariant. In the quantum regime, this maps to a closed bipartite system SE evolving unitarily. While the global von Neumann entropy remains constant ($S(\rho_{\text{SE}}) = -\mathrm{kB}\,\mathrm{Tr}(\rho\mathrm{SE}\ln\rho_{\text{SE}})$), an observer restricted to the local subsystem S tracks evolution via the traced reduced density matrix:

$$\rho_S = \mathrm{Tr}\, E(\rho\mathrm{SE}) \tag{12}$$

As system-environment entanglement develops, local subsystem purity decreases, while total information content is preserved within complex, non-local cross-correlations.

8 The Regularized Black Hole Core Hypothesis

The framework proposes a phenomenological structural volume floor at the Planck scale ($V_{\min} \sim \ell_{\mathrm{P}}^3 \approx 4.22 \times 10^{-105}$ m^3) to evaluate regular geometric alternatives to classical general-relativistic singularities. Spherically symmetric, static spacetime is modeled by the metric:

$$ds^2 = -f(r)c^2dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2 \tag{13}$$

Where the metric function is defined by a regularized radial mass function $m(r)$:

$$f(r) = 1 - \frac{2Gm(r)}{c^2 r}, \quad m(r) = M\frac{r^3}{r^3 + r_0^3} \tag{14}$$

Here, M represents the total asymptotic mass, and $r_0 = (3V_{\min}/4\pi)^{1/3}$ defines a microverse core regularization radius. At macroscopic limits ($r \gg r_0$), the mass profile approaches unity, recovering the Schwarzschild solution. At the origin ($r \to 0$), the mass function scales as $m(r) \sim r^3$, regularizing the central metric to a localized De Sitter profile: $f(r) \approx 1 - (2GM/c^2 r_0^3)r^2$. Enforcing this volume floor bounds the central mass density ($\rho_{\max} = M/V_{\min} < \infty$). For a fixed set of model parameters, this causes the central Kretschmann curvature invariant to converge to a finite central value ($K_0 = 96G^2M^2/c^4 r_0^6 < \infty$) at the coordinate origin. This finite result does not establish that the entire geometry is physically viable; the full stress-energy tensor, energy conditions, and horizon stability require separate mathematical examination.

9 The Collective Confinement and Quantum Bounce Hypotheses

The framework models the behavior of alternative quantum degrees of freedom under high-density boundaries as separate, exploratory hypotheses. Let $\mathcal{H}\mathrm{total} = \mathcal{H}\mathrm{core} \otimes \mathcal{H}\mathrm{environment}$ define a combined bipartite state space. Under closed unitary evolution, global information is conserved and total von Neumann entropy remains invariant:

$$S(\rho\mathrm{total}) = -\mathrm{kB}\,\mathrm{Tr}(\rho\mathrm{total}\ln\rho_{\mathrm{total}}) = \mathrm{constant} \tag{15}$$

As coordinates compresses toward the microverse core radius ($r \to r_0$), the model hypothesizes that individual state trajectories undergo an intense phase of dynamical confinement. Transitions between distinct configurations are suppressed by an increasing localized energy barrier, effectively freezing the observable macroscopic core dynamics while subsystem purity $\mathrm{Tr}(\rho_{\mathrm{core}}^2)$ decreases toward its mixed-state limit ($1/d_{\mathrm{core}}$), depending heavily on specific tracking track parameters and boundary couplings. Concurrently, the framework explores a quantum bounce hypothesis where this high-density boundary triggers a dynamic transition into an expanding coordinate phase.

10 The Cosmological Coupling Hypothesis

The model investigates a contested cosmological coupling track where energy distributions from regularized core arrays couple dynamically to the large-scale spacetime metric. To preserve global energy accounting, the cosmological system is modeled as a closed network governed by local covariant conservation laws: $\nabla_\mu T^{\mu\nu} = 0$. The framework explores a candidate model where the negative pressure characteristic of compressed core states exerts a scale-invariant coupling back onto the global metric. Under this formulation, the cosmic acceleration conventionally attributed to an ungrounded Dark Energy parameter is instead explored as the

macroscopic consequence of regularized core data distributions interacting with the background lattice over cosmic time. Whether this coupling mechanism aligns with observed expansion parameters without violating known exact cosmological solutions defines an independent observational track.

11 The Non-Equilibrium Relaxation and State Transition Model

The framework models rapid system relaxation as a first-order, finite-time non-equilibrium process. Let $u(t)$ represent the local volumetric energy density. Rather than utilizing a single-threshold switch, the framework enforces a state-dependent hysteresis loop configuration across two separate operational gates:

$$\text{Charging} \rightarrow \text{Reset when } u(t) \geq u_{\text{trigger}}, \quad \text{Reset} \rightarrow \text{Charging when } u(t) \leq u_{\text{restart}} \tag{16}$$

Ordered matching the rule $u_{\text{ground}} < u_{\text{restart}} < u_{\text{trigger}} < u_{\text{critical}}$. During an active reset phase ($\Theta = 0$), the density decay profile follows the differential form:

$$\frac{du}{dt} = -\frac{u - u_{\text{ground}}}{\tau_{\text{reset}}} \tag{17}$$

Where u_{ground} represents the stable baseline density of the local medium, and τ_{reset} defines a finite characteristic relaxation lifetime (s). For a fixed system control volume V, the resulting energy loss is channeled into an explicit outgoing power flux profile:

$$P_{\text{out}}(t) = -V\frac{du}{dt} = \frac{V}{\tau_{\text{reset}}}(u - u_{\text{ground}}) \tag{18}$$

Evaluating this output power at the initialization boundary of each explicit step yields an exact discrete energy balance ledger: $E_{\text{measured}} = V(u_{\text{start}} - u_{\text{discrete,end}})$.

12 The Discrete 256-Dimensional State-Space Hypothesis

The framework proposes that the underlying background informational substrate can be modeled as a discrete, finite-dimensional quantum state-space. Let $\mathcal{H}_{256}$ represent a 256-dimensional Hilbert space generated by an effective 8-qubit quantum register. Under this formulation, a general pure state vector is represented by a linear superposition of 256 computational basis states:

$$|\psi\rangle = \sum_{j=0}^{255} a_j |j\rangle, \quad \sum_{j=0}^{255} |a_j|^2 = 1 \tag{19}$$

The model explores an exploratory dimensional correspondence hypothesis where this 256-dimensional state vector maps onto specific compactified subspaces within the 24 transverse physical degrees of freedom found in light-cone configurations of bosonic string theory, configured as a discrete Leech lattice ($\mathbb{R}^{24}$). To mature this correspondence toward a formal algebraic proposal requires the future derivation of an explicit encoding map $E : \mathcal{H}256 \longrightarrow \mathcal{H}\text{lattice}$ tracking preserved inner products or information capacities under topological symmetries.

13 The Scale-Invariant Informational Mapping Postulate

The framework introduces an exploratory scaling postulate proposing that microscopic quantum-state constraints and macroscopic non-equilibrium energy density dynamics are connected through

a bounded, dimensionless scaling function. Let Z represent a dimensionless localized structural impedance parameter ($0 \leq Z \leq 1$). For a qubit subsystem, the scaling function is defined as:

$$\Lambda(Z, Q) = Q \cdot (1 - Z) \tag{20}$$

Where $Q = \mathrm{Tr}(\rho^2)$ is the formal quantum purity. To construct a mathematically bounded saturation model where energy density approaches a finite ceiling asymptotically, the framework introduces a constitutive hypothesis connecting structural impedance directly to local energy density: $Z(u) = u/u_{\text{critical}}$ for $0 \leq u \leq u_{\text{critical}}$. Injecting this function directly into the energy density balance yields the bounded time-evolution equation:

$$\frac{du}{dt} = Q\left(1 - \frac{u}{u_{\text{critical}}}\right)\left(\frac{P_{\text{in}} - P_{\text{loss}}}{V}\right) \tag{21}$$

For the following closed-form result, Q is treated as constant over the interval of interest. When purity evolves dynamically according to the coupled feedback equation, the charging solution instead depends on the time integral of $Q(s)$:

$$u(t) = u_{\text{critical}} - (u_{\text{critical}} - u_0)\exp\left[-\frac{P_{\text{in}} - P_{\text{loss}}}{V \cdot u_{\text{critical}}}\int_0^t Q(s), ds\right] \tag{22}$$

The scaling postulate represents a phenomenological approximation of energy-transfer rates; it does not independently establish spacetime regularity or gravitational equivalence, which require separate demonstrations using covariant stress-energy equations and curvature invariants. It is presented as a testable research hypothesis concerning the relationship between quantum constraints and macroscopic energy-transfer dynamics.

14 Conclusion and Future Outlook

The Dynamic Network Alignment Model provides a mathematically stable approach to simulating node attenuation and vector updates without encountering boundary crashes. By utilizing a multiplier-gated system rather than traditional division metrics, the model maintains mathematical continuity even at the absolute threshold of alignment. As an independent theoretical framework, the ultimate physical validity of these rules remains an unverified hypothesis. The accompanying computational experiments successfully verify the logical architecture programmed into the simulation engine, but they do not substitute for empirical baseline proof in natural systems. Future iterations of this research will focus on identifying potential observational anomalies or empirical datasets within distributed networks and complex systems to determine if these stable mathematical rules reflect physical parameters in nature.